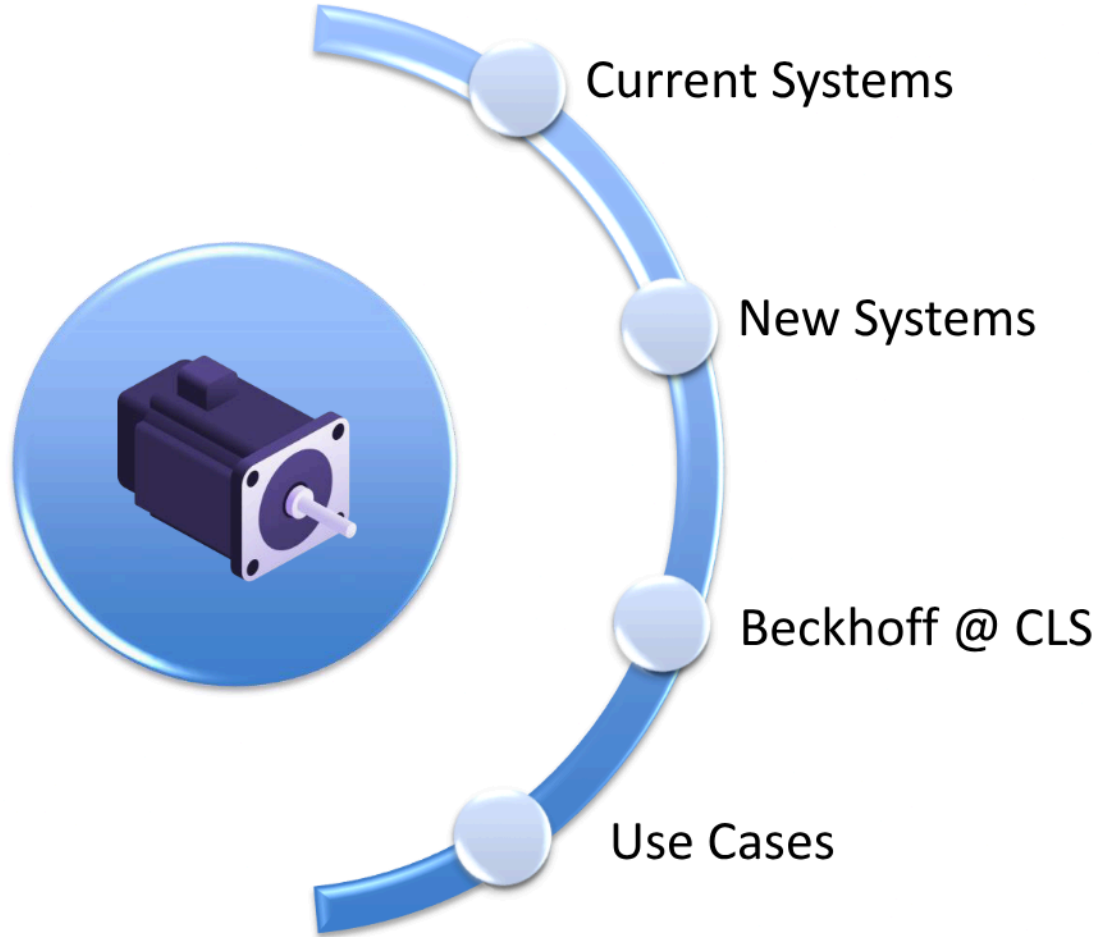


Canadian Light Source Motion Control Upgrade

Augusto Yoshio Horita

Outline



OMRON
AUTOMATION & SAFETY



DELTA TAU
Data Systems Inc.

EPICS
RTU

 **OMS**
MOTION, Inc.

BECKHOFF

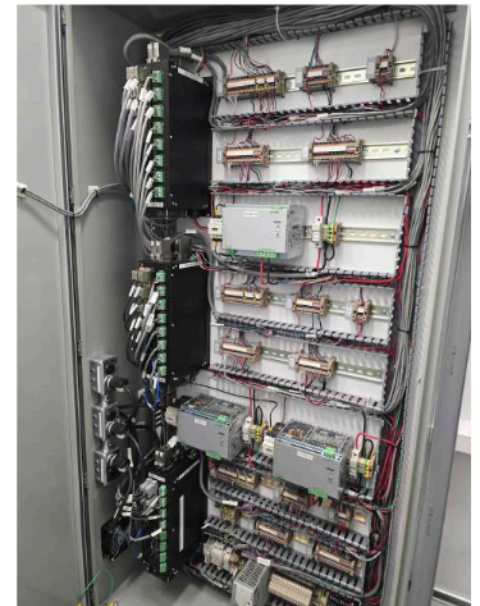


Canadian
Light
Source

Centre canadien
de rayonnement
synchrotron

Current Motion Systems

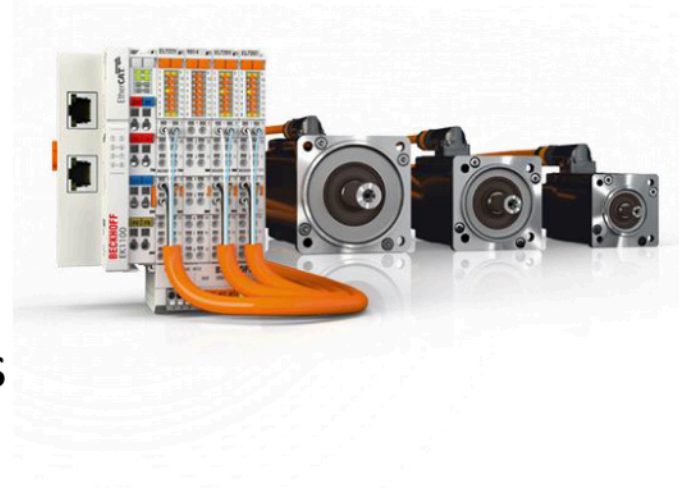
- Main standard Motion Controller
 - OMS MaxV modules – VME bus
- Specific Turn key / More complex systems
 - Omron Deltatau Motion Controllers



New Motion Systems

BECKHOFF

- Main standard Motion Controller
 - Beckhoff Motion – Ethercat
- Specific Turn key / More complex systems
 - Omron Deltatau Motion Controllers



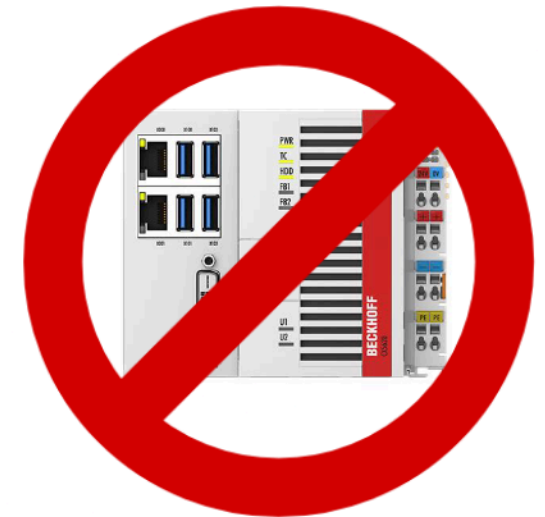
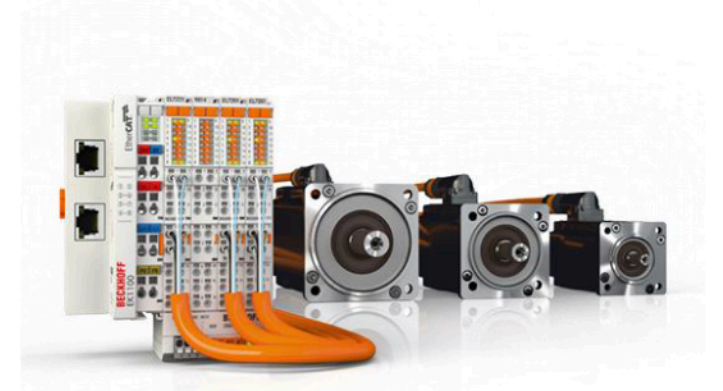
OMRON
AUTOMATION & SAFETY

 **DELTA TAU**
Data Systems Inc.



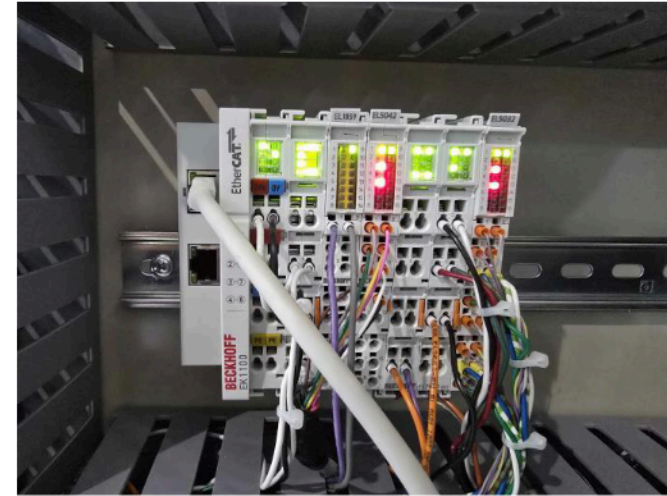
Beckhoff Motion System @ CLS

- System Architecture
 - Development framework: [ecmc](#) / [ecmccfg](#)
 - Debian 12 IOC-host
 - IOC w/ ecmc+ecmccfg (built using require)
 - Ethercat connection to Bus with coupler (EK1100)
 - Modules for motors / encoders / DIO / AIO
 - Do not use:
 - TwinCAT
 - For safety systems



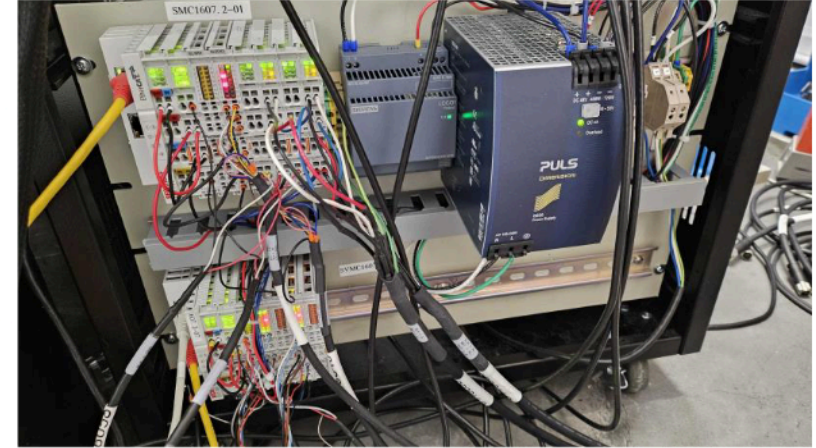
Beckhoff Motion System @ CLS

- Laboratory test with Insertion devices
 - 1 axis (gap)
 - 1 stepper motor
 - 1 absolute encoder
- Implemented and used



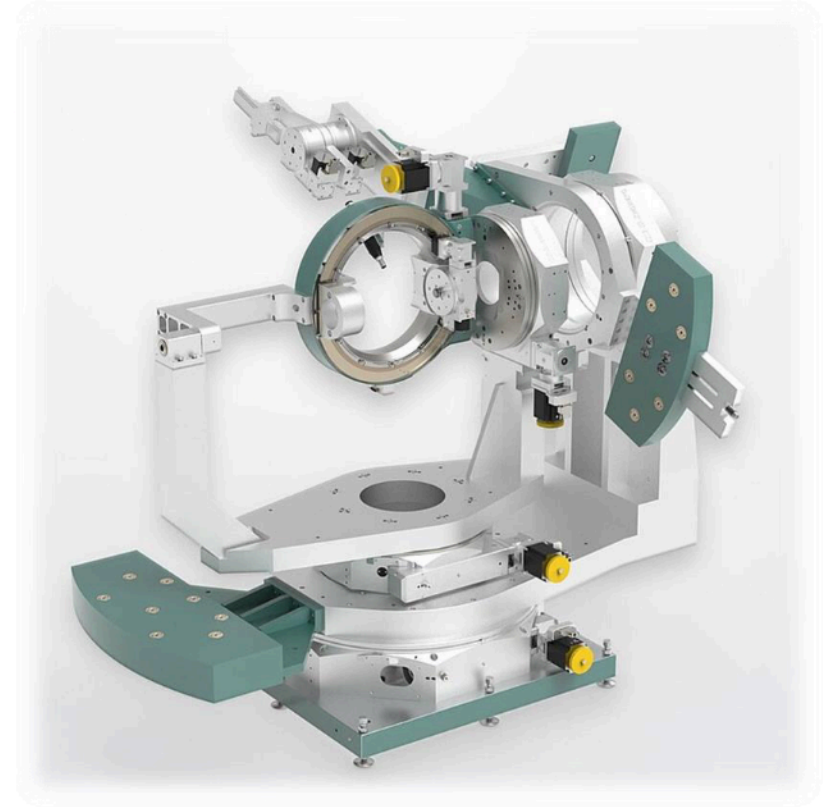
Beckhoff Motion System @ CLS

- Beamline End Station Stages
 - 2 axis each
 - Servo motors
 - Absolute encoders
- Implemented and ready to be used



Beckhoff Motion System @ CLS

- Upgrade on Huber Diffractometer
 - Replacing motion controller
 - Replacing encoders to absolute
 - Replacing some motors
- Specification / Acquisition stage





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